

## Work Experience

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| <b>Machine Learning Engineer</b><br><i>Smart Property Manager Project</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>mVizn Pte. Ltd.</b><br>Singapore (Remote)                            | <b>Jan 2024 – Present</b>  |
| <ul style="list-style-type: none"><li>Developed a <b>multi-agent</b> platform for the construction domain using <b>CrewAI</b> enabling Autonomous Project Management &amp; Scheduling (APMS), Automated Estimation &amp; Procurement (AEP), and intelligent project information retrieval increasing overall business efficiency by 50%.</li><li>Conceptualized the orchestration layer for governing through intelligent state &amp; memory management, context propagation, and dynamic routing of user intent.</li><li>Designed robust multi-agent workflows by balancing technical trade-offs and ensuring <b>model observability</b>, tracing and performance monitoring.</li><li>Built multi-modal Retrieval-Augmented Generation (<b>RAG</b>) pipelines optimizing for performance and accurate retrieval with <b>hybrid search</b> and <b>reranking</b> techniques across massive datasets of text, images and CAD files leveraging multiple types of embeddings stored in <b>Pinecone</b>.</li><li>Applied <b>LoRA</b> fine-tuning to adapt Gemini LLMs for domain understanding &amp; structured output formatting.</li></ul> |                                                                         |                            |
| <b>3D Scan-to-BIM Project</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                         |                            |
| <ul style="list-style-type: none"><li>Led the development of a scalable end-to-end machine learning pipeline from training to deployment for 3D semantic <b>segmentation</b> of indoor point clouds extracted from LiDAR scans using <b>Python</b> and <b>PyTorch</b> reducing Building Information Modeling (BIM) creation time by 60%.</li><li>Implemented <b>MLflow</b>-based experiment tracking, model versioning, model registry, and reproducible training workflows to support scalable <b>MLOps</b> practices.</li><li>Designed and implemented scalable, robust backend architectures using <b>FastAPI</b>, <b>Pydantic</b>, <b>asyncio</b> and multi threaded programming.</li><li>Optimized models through <b>ONNX</b> and deployed services using <b>Docker</b> and <b>Google Cloud Platform (GCP)</b>, including Cloud Storage, Compute Engine VMs, IAM and Cloud SQL services.</li></ul>                                                                                                                                                                                                                                 |                                                                         |                            |
| <b>Machine Learning Engineer</b><br><i>Smart Moving Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Synerjix</b><br>USA (Remote)                                         | <b>Jun 2025 – Dec 2025</b> |
| <ul style="list-style-type: none"><li>Created a Smart Moving Assistant, helping users estimate shifting cost of items using room photos.</li><li>Used <b>DinoV3</b> for object detection, <b>python</b> and <b>FastAPI</b> for backend development, <b>MySQL</b> as the Database and <b>AWS</b> (Sagemaker, EC2, EKS) as the deployment platform along with <b>CI/CD</b> and <b>Docker</b>.</li><li>Automation improved moving estimation by 70% and improved stakeholder business efficiency by 50%.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                         |                            |
| <b>Computer Vision Engineer</b><br><i>AI Assisted Neuroimaging</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Institute of Neurosciences</b><br>Kolkata, India                     | <b>Jan 2022 – Dec 2023</b> |
| <ul style="list-style-type: none"><li>Designed and implemented custom 3D Computer Vision solutions related to various Neurological disorders using <b>PyTorch</b>, <b>Python</b>, <b>OpenCV</b> and <b>Numpy</b> improving diagnosis speed by 40%.</li><li>Created custom data processing and ingestion (<b>ETL</b>) pipelines for handling massive MRI, CT images.</li><li>Converted raw patient data into valuable information with innovative data analysis and interactive dashboards for stakeholders using <b>Matplotlib</b>, <b>Streamlit</b>, <b>Docker</b>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                         |                            |
| <b>Machine Learning Engineer</b><br><i>Advanced driver-assistance system (ADAS)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Mercedes-Benz Research and Development India</b><br>Bengaluru, India | <b>Aug 2018 – Aug 2019</b> |
| <ul style="list-style-type: none"><li>Developed and optimized a computer vision pipeline for Vulnerable Road User (VRU) detection using <b>YOLOv3</b> and large-scale annotated datasets.</li><li>Designed &amp; deployed model training, evaluation, and validation workflows in <b>AWS</b>.</li><li>Built automated quality assurance (<b>QA</b>) and evaluation frameworks using Python, improving annotation throughput by 20% and data quality by 25%.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                         |                            |
| <b>Systems Engineer</b><br><i>Ultimatix Team</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Tata Consultancy Services</b><br>Kolkata, India                      | <b>Aug 2014 – Aug 2015</b> |
| <ul style="list-style-type: none"><li>Maintained and enhanced high-availability backend systems (<b>Java</b>, <b>MySQL</b>) for TCS's internal website, ensuring seamless functionality for over 100,000 users.</li><li>Streamlined feature release cycle, reducing deployment time by 20%.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                         |                            |

## Skills

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- Languages: Python, C++, SQL
- AI/Agent Systems: Multi-Agent Systems, Agent Orchestration (Crew AI, Google ADK), Tool Calling/Execution, Shared State/Memory, Production AI Systems, LangChain, LLM APIs (Gemini, Claude)
- Cloud & Ops: Google Cloud Platform (GCP), Amazon Web Services (AWS), Git, Docker
- Other: Scalable System Design, System Architecture, Workflow Orchestration

## Projects

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- **Doctor Appointment Scheduler** (Mar '26 - Apr '26) - Architected and deployed a scalable, production-level Multi-Agent System for appointment scheduling. Designed the Agent Orchestration and multi agent delegation and coordination using Google ADK, integrating intelligent doctor matching and calendar scheduling (task hand-off/execution). Implemented robust Tool Calling and Execution using Composio for calendar actions and Google Maps MCP. The scalable Backend Engineering architecture utilizes Google Cloud Run for containerized deployment, integrating Google Cloud SQL and Gemini Enterprise.  
GitHub - [https://github.com/arijitde92/Doctor\\_Appointment\\_Scheduler](https://github.com/arijitde92/Doctor_Appointment_Scheduler)  
Live Link - <https://doctor-appointment-app-173427564927.asia-south1.run.app/>
- **Job Application Agent** (Apr '25 - May '25) - Developed a real-world Multi-Agent System for automation, utilizing Crew AI for advanced workflow orchestration of specialized agents. Engineered Shared State/Memory by integrating Google BigQuery (vector store) and Vertex AI (embeddings) for consistent, scalable semantic search and storage across agents. Implemented custom tools (LinkedIn/GitHub scrapers) and leveraged LangChain for core system logic, all built in Python.  
GitHub - [https://github.com/arijitde92/Job\\_Application\\_Agent](https://github.com/arijitde92/Job_Application_Agent)

## Education

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- PhD, Jadavpur University, Kolkata, India. 2019–2024
- M.Tech. Computer Science & Engineering, Jadavpur University, Kolkata, India. GPA – 8.83 2016–2018
- B.Tech. Computer Science & Engineering, Techno India, Kolkata, India. GPA – 8.81 2010–2014

## Publications

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1. **A. De**, A. Santra, M. Tiwari and A. S. Chowdhury, "Predicting Genetic Markers for Brain Tumors Using a Composite Loss," in *IEEE Transactions on Computational Biology and Bioinformatics (TCBB)*, 2025 July. doi: 10.1109/TCBBIO.2025.3593318.
2. **A. De**, A. S. Chowdhury, (2025). Shape Induced Multi-class Deep Graph Cut for Hippocampus Subfield Segmentation. In: Antonacopoulos, A., Chaudhuri, S., Chellappa, R., Liu, CL., Bhattacharya, S., Pal, U. (eds) Pattern Recognition. (**ICPR 2024**). Lecture Notes in Computer Science, vol 15313. Springer, Cham. [https://doi.org/10.1007/978-3-031-78201-5\\_16](https://doi.org/10.1007/978-3-031-78201-5_16)
3. **A. De**, M. Tiwari, & A. S. Chowdhury. 3D Hippocampus Segmentation Using a Hog Based Loss Function with Majority Pooling. In 2023 IEEE International Conference on Image Processing (**ICIP**), Kuala Lumpur, Malaysia **2023**, October, pp. 2260–2264.
4. **A. De**, R. Mhatre, M. Tiwari and A. S. Chowdhury, "Brain Tumor Classification from Radiology and Histopathology using Deep Features and Graph Convolutional Network," 2022 26th International Conference on Pattern Recognition (**ICPR 2022**), Montreal, QC, Canada, 2022, pp. 4420–4426
5. **A. De**, M. Tiwari, E. Grisan, A.S. Chowdhury, "A Deep Graph Cut Model for 3D Brain Tumor Segmentation", 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (**EMBC 2022**).
6. **A. De**, A.S. Chowdhury, "DTI based Alzheimer's Disease Classification with Rank Modulated Fusion of CNNs and Random Forest", **Expert Syst. Appl.** 169 (2021), 114338. DOI: 10.1016/j.eswa.2020.114338

## Certifications

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- Deep Learning, a 5-course specialization, by Deeplearning.ai. Verify [here](#).
- MCPS: Microsoft Certified Professional, Microsoft Certification No.: 1042C4-5DHF30